



Market integration and price transmission in consumer markets of developing countries[☆]



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ABSTRACT

World prices for agricultural commodities surged in 2006–08, and then again in 2011–12. In many developing countries, consumer prices for staple foods, such as bread and rice, mirrored these movements. This paper examines whether prices in urban consumer markets within developing countries are co-integrated with prices in world agricultural commodity markets. Using a single equation error correction model, we examine the response of consumer prices for wheat, rice, maize, and sorghum to changes in world market prices and exchange rates in urban centers of the developing world. Analyzing over 60 country/commodity pairings, we find that developing countries' consumer markets are co-integrated with world markets. Yet, we also find that the transmission of changes in both world prices and real exchange rates to domestic consumer prices is not high, and that the movement of domestic consumer prices to new equilibrium with world prices after a shock to the latter is relatively slow.

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Introduction

World prices for agricultural commodities surged in 2006–08, and then again in 2011–12. In many developing countries, consumer prices for staple foods, such as bread and rice, mirrored these movements. The response of consumer food prices to changes in world prices – particularly for imported goods – in the urban centers of developing countries suggests a high degree of co-movement and price transmission between them.

World price volatility and high price transmission can result in countries importing world price shocks into their economies (though whether consumers and producers specifically suffer or benefit from a shock depends in part on whether prices rise or fall). The surges in world agro-food prices of recent years along with apparent high price transmission to many developing countries' domestic markets substantially reduced consumers' real income (especially where food purchases comprised a large share of households' total expenditure), drove many households into poverty, and among the poorest created or exacerbated hunger

and malnutrition.¹ It therefore would be useful to know the degree to which fluctuations in consumer food prices within developing countries are a direct result of world price movements.

This paper examines whether food prices in urban consumer markets within developing countries are co-integrated with prices in world agricultural commodity markets, as measured by the degree of co-movement and price transmission between the latter and the former. We focus on urban markets within countries because urban consumers typically depend more on imported food than do their rural counterparts.

Although empirical work on market integration and price transmission has centered more on producer rather than consumer prices (Mundlak and Larson, 1992; Quiroz and Soto, 1995; Sharma, 2003), some studies have examined the latter. Using a mix of monthly and annual consumer, wholesale, and producer agricultural prices for 16 countries (with most series ending in 2001), Conforti (2004) finds that transmission of world to domestic prices for Latin American and Asian countries is higher than for their African counterparts. Examining a subset of Sub-Saharan African countries with monthly consumer prices over a 5–10 year period, Minot (2011) also finds low transmission. On the other hand,

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¹ Although the importing of world price shocks by free-trading open economies can hurt domestic food security, closed economies are vulnerable to a different type of food insecurity – domestic supply shocks that reduce food availability can send prices soaring. FAO (2003) discusses the various types of food insecurity and their relationship to trade policies.

Cudjoe et al. (2010) determine for Ghana, and De Janvry and Sadoulet (2010) for Guatemala, that the recent price surges substantially increased consumer food prices, though the price transmission was not complete. Baquedano et al. (2011) find that although there is almost no transmission to domestic producer prices for rice in Mali, transmission to rice consumer prices is high.

The empirical scope of our paper is the examination of market co-integration and price transmission to consumer prices for wheat, rice, maize, and sorghum in the major urban centers of selected countries in Asia, Latin America and the Caribbean, and Sub-Saharan Africa during the 2000s. During the last 20–25 years, all the countries in our study have liberalized their agricultural domestic and trade policies, some quite substantially, in order to integrate into world markets and thereby capture the gains from trade based on comparative advantage. Yet, major trade impediments and distortions still exist within many countries (Anderson, 2009).

The four commodities covered by our study are major food staples within the developing world. Moreover, two of these food-stuffs – wheat and rice – are heavily imported by countries, while for the other two – maize and sorghum – domestic production supplies the bulk of domestic needs. The average import dependency of countries in our study for wheat and rice is 72% and 41%, respectively, while the import dependency of countries for maize and sorghum is 25% and 2%.

Our paper complements and expands on earlier studies in several ways. With the exception of Dawe and Slayton (2010), previous work largely ignores the effects of movement in exchange rates on domestic consumer prices in developing countries. We follow the methodology of Baquedano et al. (2011) and use a single equation error correction model, or SEECM, that separates out the effect on consumer prices of changes in world prices from changes in countries' real exchange rates. This separation is especially important in cases where a country depends heavily on imports to meet domestic demand, given that an appreciating (depreciating) currency lowers (raises) domestic prices for imported goods. We first estimate our SEECM in a panel framework, which allows us to compute and compare aggregate results (covering all countries) for each commodity. We then also compute results for each individual country/commodity pairing.

In the paper's second section, we examine factors that can affect the transmission of changes in world prices and exchange rates to a country's domestic prices. In the third and fourth sections, we present the paper's SEECM methodology and examine the data used. In the subsequent section we discuss our empirical results, and in the paper's final section we present major conclusions.

Factors affecting price transmission

The following factors can impact price transmission: (1) price and border policies; (2) changes in exchange rates; (3) market structure and conditions and transaction costs; and (4) substitutability between domestic and foreign goods. Transmission-impeding policies include systems of managed (or fixed) prices, state trading, import quotas, and trade-prohibitive tariffs (Liefert and Persaud, 2009). Exchange rate changes can also retard transmission, where the change can result from either a macroeconomic policy decision or economy-wide developments within a system of floating exchange rates. For example, the world trade price for a commodity could rise, such that if the price increase were transmitted to a country's domestic price, the latter would also rise. Assume, though, that the country's currency appreciates vis-à-vis its main trade partners for the commodity, which would have the isolated effect of lowering the domestic price for the good. The exchange rate change could thereby mitigate the trade price

increase, or even dominate it, such that the domestic price for the good falls (Dawe, 2008). In addition, weak exchange rate transmission could reduce expenditure-switching (substitution) between domestic and foreign produced goods by producers and consumers in response to world price changes (Campa and Goldberg, 2005).

Market structure and conditions that can affect price and exchange rate transmission include domestic market power and weak infrastructure. Market power gives domestic producers price-setting potential, such that changes in border prices are not transmitted completely to domestic prices. Weak infrastructure (physical, commercial, and institutional) increases transaction costs and impedes the flow of price and other key market information from borders to interior regions (Fackler and Goodwin, 2001; Barrett, 2001; Barrett and Li, 2002). This can be a particular problem within developing countries. Being landlocked could increase the infrastructure and price transmission challenges for countries, as Baquedano et al. (2011) find for Mali. Yet, whether countries are coastal or landlocked, the infrastructure servicing large urban centers is probably better than that for rural areas.

Lastly, domestic products and their foreign counterparts might not be complete substitutes (homogenous). Consequently, the transmission of world (foreign) to domestic prices for these goods will probably be incomplete.

We present transmission estimates in this paper for over 60 country/commodity pairings. Given the large number of pairings, it is infeasible to examine the possible reasons why the calculations show incomplete transmission for specific cases.

Methodology: The single equation error correction model (SEECM)

Our empirical analysis is motivated by the law of one price, which states that the price for a homogeneous commodity in two different markets should be the same, once transportation and transaction costs are adjusted for, and assuming no policy intervention. For a traded good, we can examine this causal relationship between the border price and the domestic consumer price in an error correction framework. This framework allows us to separate out and measure domestic price shocks from world markets in both the short run and long run. It also allows us to directly estimate the rate at which the domestic price returns to its equilibrium relationship (if one is shown to exist) with the world price after a change in the latter (De Boef and Keele, 2008).

We define the data generating process (DGP) for the relationship between the domestic consumer price and the world border price as:

$$p_{ijt}^d = \alpha_0 + \alpha_1 p_{ijt-1}^d + \beta_0 p_{ijt}^b + \beta_1 p_{ijt-1}^b + \varepsilon_t \quad (1)$$

where p_{ijt}^d and p_{ijt}^b are the natural log of the domestic and border price, respectively, in real terms in the domestic currency of country i of a homogeneous commodity j at time t . However, the border price equals the world price in foreign currency times the country's exchange rate. We therefore can expand the functional form in Eq. (1) by breaking p_{ijt}^b into its two parts, dropping the subscripts for country i and commodity j for simplicity:

$$p_t^d = \alpha_0 + \alpha_1 p_{t-1}^d + \phi_0 wp_t^f + \phi_1 wp_{t-1}^f + \psi_0 e_t + \psi_1 e_{t-1} + \varepsilon_t \quad (2)$$

where wp_t^f and e_t are in natural logs and represent the real world price of the commodity in foreign currency and the real exchange rate between the two countries' currencies. The coefficients ϕ_n and ψ_n measure the effects on p_t^d of an immediate and lagged change in wp_t^f and e_t , respectively.

The model in Eq. (2) represents a general form of the augmented distributed lag model, or ADL. This general form of our

model can be characterized as an ADL(1,1) process, meaning that one lag of the dependent variable and one lag of both wp_t^f and e_t are considered as regressors. The model in Eq. (2) makes no *a priori* assumptions about the appropriate lag-length of the effect of changes in both wp_t^f and e_t on p_t^d , and can be consistently estimated by OLS (De Boef and Keele, 2008; Davidson and MacKinnon, 1993). Moreover, an equilibrium relationship can be obtained from this model as long as ε_t represents a stable stationary process, or in other words, a process that is $I(0)$ (De Boef and Keele, 2008; Lütkepohl, 2005). It is from this general form of the ADL that we will develop an error correction model (ECM) that characterizes the data generating process of p_t^d , given wp_t^f and e_t .

An important feature of Eq. (2) is the causal relationship implied by the model between domestic consumer markets and the world price and exchange rate. Our model assumes causality runs from world prices and exchange rates to domestic prices. In the case where prices in two markets are endogenously determined by each other, which can occur between world markets and large exporting or importing countries which have world market power, price fluctuations in both series would be better modeled by using the approaches proposed by Johansen (1988) or Phillips (1991). These approaches rely on systems of equations in a vector auto-regressive form.

The standard modeling approach towards ECM when dealing with co-integrated time series and a unidirectional causal relationship has been the Engle and Granger (1987) two step model (henceforth abbreviated as the EGECM). In this paper, however, we rely on a single equation error correction model (SEECM), which has two distinct advantages over the EGECM. First, the SEECM does not require that all the time series being related have a unit root in order to attempt to model their long term equilibrium relationship under an ECM. As shown by Lütkepohl (2005) and Banerjee et al. (1998), requiring that all time series being related are non-stationary (which implies a unit root) to model their relationship under an ECM process mischaracterizes the nature² of these models. They show that through an ADL process—which would be the alternative to modeling stationary series—one can characterize the long term equilibrium relationship under an ECM structure between two or more time series. Second, by using an SEECM one can also derive a more robust test for co-integration and less biased parameter estimates relative to the EGECM (Banerjee et al., 1998; De Boef and Keele, 2008).

To derive the SEECM, we first subtract p_{t-1}^d from Eq. (2), which yields:

$$\Delta p_t^d = \alpha_0 + (\alpha_1 - 1)p_{t-1}^d + \phi_0 wp_t^f + \phi_1 wp_{t-1}^f + \psi_0 e_t + \psi_1 e_{t-1} + \varepsilon_t \quad (3)$$

We then add $\phi_0 wp_t^f$ to, and subtract $\psi_0 e_t$ from, the right hand side of Eq. (3), resulting in:

$$\Delta p_t^d = \alpha_0 + (\alpha_1 - 1)p_{t-1}^d + \phi_0 \Delta wp_t^f + (\phi_0 + \phi_1) wp_{t-1}^f + \psi_0 \Delta e_t + (\psi_0 + \psi_1) e_{t-1} + \varepsilon_t \quad (4)$$

Substituting and factoring common parameters, we arrive at:

$$\Delta p_t^d = \alpha_0 + \delta p_{t-1}^d + \lambda_0 \Delta wp_t^f + \lambda_1 wp_{t-1}^f + \theta_0 \Delta e_t + \theta_1 e_{t-1} + \varepsilon_t \quad (5)$$

where $\delta = (\alpha_1 - 1)$; $\lambda_0 = \phi_0$; $\lambda_1 = (\phi_0 + \phi_1)$; $\theta_0 = \psi_0$; $\theta_1 = (\psi_0 + \psi_1)$.

Rearranging Eq. (6) and collecting common terms, we derive the SEECM in Eq. (6):

² De Boef and Keele (2008) find that with few exceptions authors take this to mean that if two or more time series are stationary, they cannot be characterized by an ECM, nor can the concept of an equilibrium relationship be established. According to De Boef and Keele—based on the work of Banerjee et al. (1998)—this is not only technically incorrect but also confounds the meaning of an equilibrium relationship between two or more time series, which only requires that DGP be a stable process.

$$\Delta p_t^d = \alpha + \beta \Delta wp_t^f + \rho \Delta e_t + \delta (p_{t-1}^d - \gamma wp_{t-1}^f - \varphi e_{t-1}) + \varepsilon_t \quad (6)$$

where $\beta = \lambda_0$ and $\rho = \theta_0$ represent the short-run transmission elasticities for the world price and real exchange rate. The long run elasticity of the change in world price on domestic consumer price is given by $\gamma = \frac{\lambda_1}{\delta}$. Similarly, the long run elasticity of change in the exchange rate on the domestic consumer price is given by $\varphi = \frac{\theta_1}{\delta}$. And δ again represents the error correction term.

The long run equilibrium relationship implied by Eq. (6) between the consumer price and world price/real exchange rate is given by:

$$p_{t-1}^d - \gamma wp_{t-1}^f - \varphi e_{t-1} = 0 \quad (7)$$

This equilibrium relationship assumes that ($\gamma \neq 0, \varphi \neq 0$), and that ε_t in Eq. (6) is a stable data-generating process. However, two other long run equilibrium relationships are possible: ($\gamma = 0, \varphi \neq 0$) and ($\gamma \neq 0, \varphi = 0$). With the first condition, a long run equilibrium relationship exists between the domestic price and only the RER, while with the second condition, a relationship exists between the domestic price and only the world price.

To test for $\varphi \neq 0$, we rely on Campa and Goldberg's work (2005) on exchange rate transmission for an imported good. Following Campa and Goldberg, a value for φ not significantly different from 0 implies no transmission, while a value of 1 implies complete transmission.³ If both hypotheses are rejected, it can be said that there is partial transmission. Like their model, our model is also unconstrained, which means that $|\varphi| > 1$ is theoretically possible (though Campa and Goldberg argue that this condition is rarely observed). Again as with Campa and Goldberg, we rely on Fisher's transformation⁴ of the parameter estimates to constrain them to 1 before proceeding to test the null.

Before moving to the topic of the adjustment to equilibrium of consumer prices after a world price (or exchange rate) shock, we briefly discuss the differences between the tests for a long run relationship for the EGECM and SEECM. This will provide a better understanding as to how the latter approach results in more robust tests and parameter estimates. In its test for a long run relationship, the EGECM fits the model in Eq. (8) as a first step regression:

$$p_t^d - \phi_0 wp_t^f - \psi_0 e_t = 0 \quad (8)$$

Eq. (8) is fitted only after determining whether or not the series being related are of the same order of integration $I(b)$, and whether the null of non-stationarity of the series cannot be rejected using one of several variants of the Dickey Fuller (DF) or Phillip-Perron (PP) unit root tests. The EGECM rejects the null of no long run relationship with either the world price or the RER when $(\phi_0, \psi_0) \neq 0$ and ε_t is stationary. However, upon closer inspection of Eq. (8), one can see that it represents a restricted form of the general ADL in Eq. (2) and imposes the following condition on its lagged terms: $\alpha_1 = \phi_1 = \psi_1 = 0$. This condition implies an instant adjustment of the domestic price to a change in the import (world) price or the real exchange rate. If domestic prices do not adjust instantly, as we have argued in the second section of this paper, both ϕ_0 and ψ_0 are potentially biased, given that the true test of a long run relationship should be based on the parameters (γ, φ)⁵ derived from ADL.

³ Campa and Goldberg's derivation of these values comes from the definition of the true exchange rate effect, which according to their work is $(1 + \Phi)$ with $\Phi = 0$ ($\Phi = -1$). According to their derivation, $(1 + \Phi)$ allows one to control for the structure of competition in a particular industry, which includes macro-economic conditions within the importing country. Their definition of the true exchange rate effect implies values of $\varphi = 1$ ($\varphi = 0$), that is, complete/no transmission.

⁴ The transformed parameter estimate under Fisher is given by $z = 1/2[\ln(1 + \hat{\varphi}) - \ln(1 - \hat{\varphi})]$. Campa and Goldberg state that this transformation tends to reject equality to 0 or 1 a bit more frequently.

⁵ For a measure of the bias, see footnote 7 in De Boef and Keele (2008).

The bias affects not only the test for a long run relationship, but also the estimate of the error correction term—or ECT—in the second step regression of the EGECM, given that it is derived from the biased ε_t in Eq. (8). In both the EGECM and SEECM, δ represents the ECT. This term measures the speed at which the domestic consumer price (p_t^d) returns to its long run equilibrium relationship with the world price or exchange rate after a shock in either of these latter two variables. Its sign is expected to be negative, as it shrinks the gap between the series in each subsequent period. $|\delta|$ can vary between the values of 0 and 1. A value of 0 means that there is no long run equilibrium relationship between the domestic price and/or the world price and exchange rate, while a value of 1 means that the domestic price moves immediately to a new equilibrium given a shock in the world price or real exchange rate.

If $|\delta| \neq 0$ and one of the equilibrium relationships in the SEECM holds, as discussed in Eq. (7), it can be said that the series are co-integrated. The co-integrating test is based on the t -test on the estimated parameters. The power of this test not only is significantly higher than the Dickey Fuller test for co-integration, but it also allows one to test data that are either stationary or near-integrated, as opposed to only data series that are an $I(1)$ (Banerjee et al., 1998; De Boef and Granato, 1999).⁶

In the EGECM, however, co-integration is based not only on $|\delta| \neq 0$ but also on ε_t of Eq. (8) being stationary. Yet as discussed earlier, the error term in the static regression is most likely biased. The SEECM derived from a more flexible model, such as the ADL, allows one to test simultaneously $|\delta| \neq 0$ and $\varepsilon_t = I(0)$ to establish co-integration. This reduces⁷ or eliminates the likelihood of biased parameter estimates, and thus is a more robust test for co-integration.

To better understand the adjustment process back to equilibrium (as determined by the ECM) for the domestic consumer price in response to a shock in the world price, we give an example. We assume values of $|\delta| = 0.2$, $\gamma = 0.5$, and $\varphi = 0$, which respectively are the parameter on the ECT, the long run transmission coefficient for the world price, and the long run transmission coefficient for the real exchange rate. Adjustment of the domestic price to long run equilibrium occurs over a number of periods. $|\delta|$ Specifically measures how much of the gap between the world and domestic price inherited from the previous period is corrected (eliminated) through a domestic price change in each period (Quiroz and Soto, 1995). γ in Eq. (6) is the long run elasticity (LRE) of the effect of change in the world price on the consumer price, as opposed to the short run elasticity (as captured in Eq. (6) by β).

A value of $\gamma = 0.5$ means that in the long run, a 100% rise in the world price would increase the domestic price by 50%. (This increase is in addition to any possible short term rise in the domestic price that might occur, which would happen if $\beta \neq 0$.) However, given that $|\delta| = 0.2$, in the immediate period of the long run adjustment ($r = 0$), p_t^d would increase by only 10%, as determined by $50 * 0.2 = 10$. In the following period ($r = 1$), the domestic price would increase by an additional 8% from its initial pre-shock value ($40 * 0.2$), for a total price rise of 18%. By $r = 2$, p_t^d would increase in total by 24% from its initial pre-shock value (rising by an additional $32 * 0.2 = 6$). By $r = 3$, p_t^d would increase in total by 29% (an additional $26 * 0.2 = 5$). This adjustment process would continue until the 50% increase in p_t^d is fully attained.

By providing a measure of the long run adjustment process for a shock in the world price (or RER) on the consumer price, the SEECM also allows us to measure the median lag-length of the shock's effect. We define the median lag-length as the value of r (number of periods) at which at least half of the new equilibrium value of the domestic price from the world price shock is reached. In our example with $|\delta| = 0.2$ and $\gamma = 0.5$, the median lag-length occurs when $r = 3$.

As shown by De Boef and Keele, we can express the median lag-length mathematically as:

$$m = \frac{\sum_{r=0}^R v_r}{\sum_{r=0}^{\infty} v_r} \quad (9)$$

where $\sum_{r=0}^{\infty} v_r$ is the long run transmission elasticity of the world price γ . The numerator $\sum_{r=0}^R v_r$ is the summation across any number of periods R of the adjustment of the consumer price toward its long run equilibrium value (as determined by γ). Thus, the median lag-length of the long run adjustment process between the consumer and world price is obtained when $m \geq 0.50$. Eq. (9) also allows us to characterize and develop the adjustment process from a world price shock, since we can calculate v_r for any lag period r .

To simultaneously estimate the long run elasticities and standard errors for both wp_t^f and e_t in Eq. (6), we rely on the ADL specified in Eq. (5) and on Bewley's (1979) transformation. In the appendix, we examine this transformation in greater detail. However, as De Boef and Keele (2008) state based on Bewley, the transformation is a computational convenience for calculating the long run elasticities and their standard errors, and is not meant to represent the underlying dynamics of the DGP.

We first estimate Eq. (6) using a panel approach, and controlling for country fixed effects, to determine the degree of long run world price/RER transmission and co-integration for domestic markets/prices in the developing world for maize, rice, sorghum, and wheat. We assume the error term of this model follows a first-order autoregressive process.

$$\Delta p_{it}^d = \alpha + v_i + \bar{\beta} \Delta wp_{it}^f + \rho \Delta e_{it} + \bar{\delta} (p_{it-1}^d - \bar{\gamma} wp_{it}^f - \varphi e_{it}) + \epsilon_{it} \quad (10)$$

where $\varepsilon_{it} = \sigma \varepsilon_{it-1} + \eta_{it-1}$, $|\sigma| < 1$, and η_{it-1} is iid with mean zero and variance λ_{η}^2 . v_i represents country fixed parameters that may or may not be correlated with the covariates. If needed to further correct for serial autocorrelation, first differenced short run lags for the world price, exchange rate, or dependent variables were added. To determine appropriate lag-lengths, we relied on the Akaike information criterion (AIC).⁸ The model is estimated using Stata 12 with the *xtregar* estimation procedure.

We then fit Eq. (6) for each country commodity pair. Once again, the errors are assumed to follow a first-order autoregressive process. Our estimation is carried out using the *prais* estimation command with robust standard errors in Stata 12. The *prais* estimation command, or Prais–Winsten and Cochrane–Orcutt regression, relies on the generalized least-squares method to calculate the coefficients of a linear regression model whose errors are serially correlated. Similarly to the panel approach, we added contemporaneous lags when necessary and used the AIC to determine their length.

⁶ According to De Boef and Granato (1999), near-integrated data are series that are $I(1 - c)$, with c representing a very small number such that the order of integration is just below one. DeBoef and Granato also show that the co-integration test based on the EGECM often leads to erroneous conclusions, because of its lower power. In particular, the test rejects co-integration too easily, especially when dealing with near-integrated data.

⁷ Whether or not biased parameter estimates are obtained in the SEECM depends on whether the specified model is a good approximation of the true data-generating process.

⁸ In our modeling approach, we begin with an ADL (8,8;2), which implies eight lags for both the dependent and independent contemporaneous variables. From this structure, we eliminate insignificant lags to arrive at a more parsimonious model taking into account the AIC and values for the Durbin–Watson statistic. All models were estimated with robust standard errors.

Data and Estimation Bias

The period of estimation varies a bit by country/commodity pairing, but for all it falls within the 2000s, and for most it covers at least 5 years, including the years of the recent world price surges (up to May 2011). The consumer prices we use for wheat, rice, maize, and sorghum are from FAO's Global Information and Early Warning System (GIEWS) database and the Famine Early Warning Systems Network (FEWSNET). All prices are monthly per metric ton in the capital city of the countries listed in Tables 1 and 2. The prices have been converted to real values using the countries' domestic consumer price index (CPI), obtained from the IMF's International Financial Statistics. World prices for these commodities are from USDA and the International Monetary Fund's commodity price database, converted to real values using the U.S. CPI, and exchange rates are also from the IMF. We compute real exchange rates conventionally as the nominal exchange rate times the ratio of foreign to domestic price indices, again using the U.S. CPI for foreign prices.

For a number of reasons (some related to data), our transmission estimates understate the degree of actual transmission. The first reason is because we are computing transmission elasticities specifically for consumer food prices. Correcting for this problem requires subtracting from the prices the domestic processing and retail costs, as well as any other domestic margins, which are non-traded value-added, in order to make the domestic prices comparable with the goods' import prices. For example, in our empirical work the traded commodity is wheat, but the corresponding consumer good is the processed product of wheat flour (or bread).

Another potential source of transmission estimate inaccuracy is that the imported product and its closest domestic equivalent might differ in quality, or some other way that makes the imported and domestic good nonhomogeneous. If the imported good is of higher (lower) quality than its domestic analog, then the transmission elasticity estimate is understated (overstated). Because our study covers developing countries whose domestically produced foodstuffs are often of lower quality than imports, our transmission estimates with respect to this issue are generally understated.

Understatement also occurs because we do not adjust the domestic prices for domestic transport and transaction costs (TT). The data necessary for these adjustments are not available and would be painstaking to obtain for all our country/commodity pairings. Adjusting for TT to avoid estimate understatement is required whether one is computing transmission elasticities for domestic consumer or producer prices. For a net import, this correction involves subtracting the TT from the domestic price (as is relevant for the empirical work in this paper). Failure to make this adjustment understates our transmission estimates (Sharma, 2003).⁹

⁹ If a good is a net export, failure to adjust for TT while estimating transmission elasticities would generate overstated estimates. In the case of an imported good, the argument that TT results in an understated estimate of the "real" transmission elasticity rests on the idea that the TT makes the imported good at its domestic consumption/use site different from when it arrives at the border, because in domestic transit the good has incurred TT. The TT increases the domestic price above the border price. The border and domestically consumed goods are no longer identical in a cost/economic sense, thereby creating the same type of problem in measuring transmission elasticity as when the imported and domestically consumed analog differ in quality. In the case of TT, however, one could alternatively argue that because the imported good is physically the same at both the border and its domestic consumption/use site, the calculated transmission elasticity does not understate the real elasticity. Rather, because of the TT, the real elasticity should be understood to have a value less than 1, even if complete transmission exists.

Results

As discussed in the previous section, our model is not constrained by the *a priori* condition that the data series be non-stationary in order to evaluate the long run relationship of consumer prices to world prices. We therefore do not spend much time discussing the stationary properties of the different data series used. Nonetheless, to evaluate the stationary properties of our data series for both world prices and exchange rates, we use the GLS version of the Dickey–Fuller test (DF–GLS) and the Phillips–Perron (PP) test.¹⁰ Only in a few cases can we reject the null of non-stationarity at a 5% or higher level of significance, and in even fewer cases can we reject the null in both the DF–GLS and PP tests (Tables 1 and 2). These results imply that for the vast majority of our sample, we could have used the EG two step approach. However, as we discussed previously, that approach has the potential to generate results that bias not only our coefficient estimates but our hypotheses tests as well. The SEECM approach that we use instead not only corrects these problems, but also has the benefit of allowing us to model the long run relationship between world and consumer prices for the country/commodity pairings for which we can reject non-stationarity.

Our aggregate results for wheat, rice, maize, and sorghum show that across our sample of developing countries, consumer food prices have a long run co-integrating relationship with world prices (Table 3). For all four products, both the parameter on the error correction term and the world price are significantly different from zero. The results thereby show that consumer prices in the developing world are integrated with world prices. However, the results indicate that the transmission of changes in world prices to domestic consumer prices is not high. The average of the estimated long run transmission elasticities for the world to domestic consumer price for the four products is 0.25 (Table 3). Wheat has the highest world price transmission elasticity at 0.32%, and sorghum the lowest at 0.16%.

As discussed earlier in the methodology section, in addition to testing for co-integration, our model allows us to calculate the long run adjustment process and movement to a new equilibrium for consumer prices in response to a shock in world prices. This allows us to gauge how persistent a world price shock is on domestic consumer prices. The results show that the movement of domestic consumer prices to new equilibrium with world prices is fairly slow. Recall that the error correction term δ can vary between 0 and |1|, with a value of 0 generating no movement to new equilibrium and implying no co-integration, while a value of |1| implies that domestic prices move immediately to equilibrium with the world price or exchange rate. The average value of the estimated error correction terms for the four aggregate products in Table 3 is a low 0.15.

Earlier, we defined the median lag-length of a world price shock adjustment as the period of time it takes the domestic price to reach half of its new equilibrium value after the world price shock. Of the four commodities in Table 3, the fastest rates of adjustment (as measured by the median lag-length) are for maize and rice, at 2 and 3 months, respectively, followed by sorghum at 4 months (Fig. 1). Wheat, the most heavily imported commodity, adjusts the slowest, with a median lag-length of 7 months.

Concerning the real exchange rate, we find that all the parameter coefficients are significant. This indicates co-movement with

¹⁰ Both the DF–GLS and PP tests were done using a constant with and without a deterministic trend. Because of space constraints, we present only the results *without trend*, as they do not change our conclusions about the stationarity properties of our data series. The tests were initially conducted with a maximum of 8 lags. However, the tests results were no different when using a more parsimonious lag structure of two lags.

Table 1

Unit root test for rice and wheat price series. Source: Authors calculations.

	Consumer price		World price		Exchange rate	
	DF-GLS	PP	DF-GLS	PP	DF-GLS	PP
<i>Rice</i>						
Senegal	−0.98	−1.86	−1.87	−1.92	−0.96	−1.02
Mozambique	−0.58	−0.64	−0.64	−1.31	−0.57	−1.77
Niger	−2.56**	−2.39**	−1.12	−0.98	−0.34	−0.67
Honduras	−0.62	−0.56	−1.41	−1.68	−1.78*	0.42
Cameroon	−0.52	−2.10	−1.65*	−1.60	−1.78*	−1.60
Benin	−1.31	−1.91	−1.12	−0.98	−0.20	−0.86
Haiti	−2.63***	−2.17	−1.69*	−1.55	−0.01	−0.98
El Salvador	−1.61	−1.42	−0.64	−0.96	1.04	0.13
Guatemala	−0.82	−1.17	−1.27	−2.22	−0.63	−1.84
Costa Rica	−0.55	−1.36	−1.47	−1.97	−0.45	1.14
Nicaragua	−0.70	−1.03	−1.19	−0.98	−0.97	−1.19
Mali	−1.47	−2.34	−0.76	−1.75	−1.16	−1.74
Philippines	−2.04**	−2.14	−1.08	−0.93	−0.24	0.19
Dominican Republic	−1.76*	−1.60	−1.54	−1.82	−1.90*	−2.24
Nepal	−1.36	−2.15	−1.65*	−1.57	0.98	−0.39
Malawi	−0.99	−2.34	−0.68	−1.82	0.87	−2.78*
Peru	−1.33	−1.80	−1.87*	−1.71	0.82	0.04
Bangladesh	−1.67*	−1.91	−1.55	−1.70	−0.07	−0.71
Chad	−4.14***	−4.22***	−0.69	−1.34	−2.04**	−2.46
Pakistan	−1.51	−1.66	−1.56	−1.81	0.50	−0.50
India	−0.87	−2.07	−1.18	−1.01	0.70	0.32
Vietnam	−3.36***	−2.34	−1.85*	−1.97	−0.12	−1.80
Ecuador	−1.47	−1.41	−1.4739	−1.63	0.18	0.20
<i>Wheat</i>						
El Salvador	−0.49	−1.27	−1.38	−1.79	0.25	−1.10
Sri Lanka	−0.64	−2.42	−1.79*	−1.63	0.16	−1.28
Haiti	−2.27**	−2.24	−1.98	−1.77	−1.70	−1.50
Cameroon	−2.46**	−2.62*	−1.26	−1.55	−1.78*	−1.60
Ecuador	−1.38	−1.30	−1.25	−1.54	0.18	0.20
Peru	−2.81***	−1.79	−0.81	−1.66	0.82	0.04
Bangladesh	−1.67*	−1.46	−1.11	−1.87	−0.07	−0.71
Ethiopia	−1.00	−2.17	−1.26	−1.55	−0.78	−1.66
India	−2.25**	−1.86	−0.60	−1.50	0.70	0.32
Pakistan	−1.11	−1.33	−1.40	−1.77	0.50	−0.50
Nepal	−2.39**	−2.38	−1.26	−1.55	0.98	−0.39

Notes: DF-GLS: Dickey–Fuller GLS unit root test; PP: Phillips–Perron unit root test. DFGLS critical values: *** (Pr ≤ 0.01) = −2.59; ** (Pr ≤ 0.05) = −1.94; * (Pr ≤ 0.10) = −1.61. PP critical values: *** (Pr ≤ 0.01) = −3.45; ** (Pr ≤ 0.05) = −2.88; * (Pr ≤ 0.10) = −2.58.

consumer prices (Table 3). What about the transmission of RER changes to domestic prices? As discussed earlier in the methodology section, a null hypothesis of $\varphi = 0$ implies no transmission, while complete transmission occurs when we cannot reject the null of $\varphi = 1$. To test these two hypotheses, we follow the procedure of Campa and Goldberg.

In Table 3, we reject both the no and complete transmission hypotheses concerning the real exchange rate for all four products. This indicates that partial transmission exists. Moreover, the estimated (real) exchange rate elasticities are low. For wheat, maize, and sorghum, the average elasticity is 0.12, while for rice the estimated elasticity is a negative 0.12. Our results are generally consistent with Gilbert (2010), who investigated the effect of changes in the exchange rate of the US dollar on world commodity prices. He found that although the effects of exchange rates changes were statistically significant, the magnitude of the effects on prices was small.

Tables 4 and 5 present our results for the specific country/commodity pairings. The results again show that in general a long run relationship exists between world prices and domestic consumer prices. Based on statistical significance (for both the error correction term and world price), all 11 of the country/commodity pairings for wheat and 21 of the 23 pairings for rice indicate a long relationship between the two prices. For maize this results holds for 14 of 19 countries, and for sorghum for 5 of 8 countries. Our finding that market integration is more common for the more heavily traded commodities of wheat

and rice than for the less traded products of maize and sorghum is consistent with Minot (2011), who uses an approach similar to ours to examine price transmission in Sub-Saharan African countries. He finds that the domestic prices for rice have a long run relationship with world prices more often than do domestic prices for maize.

Our results are also consistent with Quiroz and Soto (1995). Using annual data that generally cover the 30 or so year period from the mid 1960's to the mid 1990's, they find a co-integrating relationship between world and domestic prices for the majority of countries in their study (48 out of 78).¹¹ Of the 48 countries for which Quiroz and Soto find price co-integration, 13 are included in our country coverage, for at least one of the four grain products – Burkina Faso, Cameroon, Costa Rica, Dominican Republic, Ecuador, El Salvador, Ethiopia, Honduras, Kenya, Malawi, Niger, Pakistan, Malawi, and Niger. For each of these 13 cases, we also find statistical significance for market integration. Of the 30 countries for which Quiroz and Soto do not find market integration, the only country that we cover is Bangladesh. Although our study examines a more recent time period than Quiroz and Soto, we also find no co-integrating price relationship for Bangladesh rice, though we do for wheat.

¹¹ Comparing our results to Quiroz and Soto is challenging, because they work with annual price data as opposed to our monthly data, and they calculate country-wide results that aggregate across commodities. In contrast, our study computes results for each of four grain products that aggregate across countries, as well as results for many individual country/commodity pairings.

Table 2

Unit root test for maize and sorghum price series. Source: Authors calculations.

	Consumer price		World price		Exchange rate	
	DF-GLS	PP	DF-GLS	PP	DF-GLS	PP
<i>Maize</i>						
Dominican Republic	−2.94***	−3.62***	0.08	−1.54	−1.90*	−2.24
Niger	−1.32	−2.31	0.24	−0.85	−0.57	−1.64
Peru	−1.01	−1.18	−0.40	−0.68	0.82	0.04
El Salvador	−1.52	−1.62	0.20	−0.96	1.04	0.13
Honduras	−1.39	−1.94	−0.44	−0.87	2.45**	1.30
Ecuador	0.10	−2.41	−2.03	−2.44	0.18	0.20
Guatemala	−0.87	−0.93	0.19	−1.32	−0.63	−1.84
Nicaragua	−2.94***	−3.77***	−0.64	−0.82	−0.97	−1.19
Mozambique	−2.13**	−2.95**	−0.27	−0.94	−0.57	−1.77
Chad	−3.43***	−3.56***	−0.21	−0.86	−2.04**	−2.46
Zambia	−0.45	−2.37	−0.27	−0.94	0.08	−1.24
Haiti	0.22	−0.64	0.26	−1.08	0.05	−0.98
Kenya	−1.51	−1.72	−0.34	−0.80	0.20	−0.87
Philippines	−2.10**	−2.32	−0.52	−0.78	−0.24	0.19
Malawi	−2.20**	−2.31	−0.05	−0.84	0.87	−2.78*
Mali	−1.62*	−2.57	0.18	−0.80	−1.16	−1.74
Ethiopia	−1.52	−2.16	0.06	−0.75	−0.47	−1.62
Burkina Faso	−2.15**	−2.42	0.25	−0.84	−0.63	−1.82
Benin	−3.20***	−2.62*	−0.44	−0.69	−0.20	−0.86
<i>Sorghum</i>						
Benin	−2.18**	−3.23**	−0.47	−1.42	−0.20	−0.86
Chad	−2.36**	−2.43	−0.45	−1.45	−2.04**	−2.46
Niger	−1.71*	2.52	−0.55	−1.31	−0.40	−1.50
Senegal	1.75*	−1.62	−1.21	−1.52	−0.96	−1.02
El Salvador	−1.61	−2.71*	−0.03	−1.67	1.04	0.14
Ethiopia	−1.52	−2.79*	−0.17	−1.53	−0.78	−1.66
Mali	−2.84***	−2.30	−0.65	−1.26	−1.16	−1.74
Burkina Faso	−2.32**	−2.28	−0.55	−1.31	−0.63	−1.82

Notes: DF-GLS: Dickey–Fuller GLS unit root test; PP: Phillips–Perron unit root test. DFGLS critical values: *** ($\Pr \leq 0.01$) = −2.59; ** ($\Pr \leq 0.05$) = −1.94; * ($\Pr \leq 0.10$) = −1.61. PP critical values: *** ($\Pr \leq 0.01$) = −3.45; ** ($\Pr \leq 0.05$) = −2.88; * ($\Pr \leq 0.10$) = −2.58.

Table 3

Long run world price and exchange rate elasticities by commodity. Source: Authors calculations.

	Error correction term	World price	Real exchange rate	No exchange rate transmission	Complete exchange rate transmission	Median lag length	Import dependency (%)	Observations	Adj. R2	DW
Wheat	−0.09*** (0.015)	0.32*** (0.008)	0.10*** (0.010)	†	‡	7	72	848	0.06	1.82
Rice	−0.16*** (0.014)	0.24*** (0.004)	−0.12*** (0.008)	†	‡	3	41	2,103	0.16	2.00
Maize	−0.22*** (0.017)	0.30*** (0.014)	0.12*** (0.017)	†	‡	2	25	1,874	0.14	1.97
Sorghum	−0.13*** (0.018)	0.16*** (0.013)	0.15*** (0.020)	†	‡	4	2	768	0.06	1.88

Note: Regression estimates are obtained from a panel that controls for country fixed effects. Robust standard errors are shown in parentheses below the parameter estimates. Durbin Watson statistic is based on the Bhargava, Franzini, and Narendranathan (1982) transformation. R2 is the within estimate. *** ($\Pr \leq 0.01$). † means $\varphi = 0$ can be rejected at the 5% significance level; ‡ means $\varphi = 1$ can be rejected at the 5% significance level.

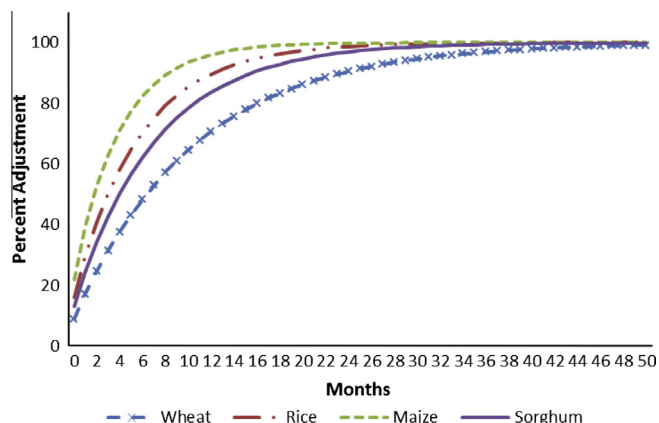


Fig. 1. Percent adjustment of consumer prices to a 1% shock in the world price by commodity. Source: Authors calculations from regression results.

Conforti (2004) also covers an earlier period than our study (most of his price series end by 2001), and his country coverage includes only two¹² countries that we also cover – Ethiopia and Senegal. For Ethiopia, both Conforti and we find a co-integrating price relationship for wheat and sorghum, while for maize, Conforti finds market integration and we do not. For Senegal, the only shared product is rice, for which both Conforti and we find co-integration.

Although our results show a long run relationship between world and domestic consumer prices (in terms of statistical significance), they do not indicate high price transmission nor fast adjustment of domestic prices to world price shocks. The number of countries with a transmission elasticity greater than 0.50% are 6 of 23 for rice, 1 of 11 for wheat, 2 of 14 for maize, and 0 of 5

¹² Conforti also uses annual rather than monthly data, and focuses on exported rather than imported products. That is why there is so little overlap between his and our country coverage.

Table 4Long run world price and exchange rate elasticities for rice and wheat by country. *Source:* Authors calculations.

	Error correction term	World price	Real exchange rate	No exchange rate transmission	Complete exchange rate transmission	Median lag length	Import dependency (%)	Observations	Adj. R2	DW
<i>Rice</i>										
Senegal	−0.60*** (0.158)	0.76*** (0.069)	−0.04 (0.147)		‡	<1	95	49	0.60	1.91
Mozambique	−0.22*** (0.040)	0.38*** (0.013)	−0.21*** (0.015)	†	‡	2	93	109	0.29	2.02
Niger	−0.43*** (0.069)	0.26*** (0.013)	−0.25*** (0.027)	†	‡	1	89	133	0.23	1.99
Honduras	−0.20*** (0.045)	0.72*** (0.019)	1.11*** (0.053)			2	88	101	0.21	2.03
Benin	−0.10 (0.087)	0.35*** (0.012)	−0.49*** (0.022)	†	‡		86	136	0.03	2.05
Cameroon	−0.14** (0.061)	0.35*** (0.011)	−0.52*** (0.054)	†	‡	4	86	75	0.18	2.05
Haiti	−0.30*** (0.078)	0.55*** (0.044)	0.54*** (0.099)	†	‡	1	85	74	0.41	2.04
El Salvador	−0.04** (0.017)	0.67*** (0.019)	3.41*** (0.183)			16	79	127	0.19	2.06
Guatemala	−0.13*** (0.049)	0.62*** (0.020)	−0.81*** (0.030)	†	‡	4	77	65	0.58	2.03
Costa Rica	−0.15** (0.065)	0.27*** (0.018)	−0.21*** (0.010)	†	‡	4	50	92	0.39	2.01
Nicaragua	−0.71*** (0.132)	0.42*** (0.046)	−0.03 (0.052)		‡	<1	41	131	0.36	2.09
Mali	−0.31*** (0.110)	0.20*** (0.006)	−0.15*** (0.036)	†	‡	1	29	87	0.41	1.97
Philippines	−0.15*** (0.057)	0.22*** (0.006)	0.53*** (0.026)	†	‡	4	14	132	0.47	1.98
Dominican Rep.	−0.25*** (0.054)	0.39*** (0.019)	0.52*** (0.046)	†	‡	2	6	60	0.53	2.09
Nepal	−0.18* (0.102)	0.35*** (0.025)	0.44*** (0.064)	†	‡	3	5.6	75	0.12	2.01
Malawi	−0.15*** (0.045)	0.40*** (0.022)	0.01 (0.029)		‡	4	5	84	0.55	2.08
Peru	−0.03*** (0.010)	0.12*** (0.006)	−0.59*** (0.022)	†	‡	22	4	131	0.52	2.02
Bangladesh	−0.09** (0.045)	−0.02 (0.027)	−0.34*** (0.055)	†	‡		2.8	67	0.22	1.99
Chad	−0.33** (0.14)	0.16*** (0.021)	0.42*** (0.080)	†	‡	1	3	108	0.37	2.01
India	−0.25** (0.104)	0.25*** (0.019)	0.29*** (0.058)	†	‡	2	<1	131	0.35	1.99
Vietnam	−0.61*** (0.138)	0.15*** (0.031)	0.43*** (0.042)	†	‡	<1	<1	49	0.55	2.06
Pakistan	−0.10*** (0.023)	0.53*** (0.019)	0.12*** (0.015)	†	‡	6	<1	60	0.75	2.07
Ecuador	−0.09* (0.054)	0.24*** (0.010)	0.38*** (0.066)	†	‡	7	0	67	0.11	2.02
<i>Wheat</i>										
El Salvador	−0.19** (0.075)	0.24*** (0.030)	−5.95*** (0.296)			3	164	63	0.20	2.04
Sri Lanka	−0.38** (0.144)	0.31*** (0.023)	−0.39*** (0.045)	†	‡	<1	117	54	0.25	2.11
Haiti	−0.16*** (0.058)	0.32*** (0.027)	0.17*** (0.049)	†	‡	3	100	74	0.41	2.00
Cameroon	−0.79*** (0.102)	0.34* (0.184)	−0.60** (0.274)		‡	<1	100	71	0.59	1.77
Ecuador	−0.26* (0.139)	0.48*** (0.032)	−1.64*** (0.134)			2	99	67	0.15	1.97
Peru	−0.03** (0.013)	0.24*** (0.004)	−0.15*** (0.010)	†	‡	22	94	131	0.63	2.03
Bangladesh	−0.05* (0.028)	0.32*** (0.022)	1.68*** (0.040)			12	85	65	0.77	2.04
Ethiopia	−0.39*** (0.120)	0.73*** (0.081)	0.12*** (0.033)	†	‡	1	25	72	0.53	1.91
India	−0.06** (0.024)	0.31*** (0.006)	0.19*** (0.014)	†	‡	10	4	131	0.51	1.955
Pakistan	−0.27** (0.108)	0.37*** (0.083)	0.10 (0.062)		‡	<2	4	60	0.64	1.961
Nepal	−0.13* (0.071)	0.12*** (0.019)	0.12*** (0.023)	†	‡	4	1	73	0.30	2.05

Note: Robust standard errors are shown in parentheses below the parameter estimates. *** (Pr ≤ 0.01); ** (Pr ≤ 0.05); * (Pr ≤ 0.10). † means $\varphi = 0$ can be rejected at the 5% significance level; ‡ means $\varphi = 1$ can be rejected at the 5% significance level.

Table 5

Long run world price and exchange rate elasticity for maize and sorghum by country. Source: Authors calculations.

	Error correction term	World price	Real exchange rate	No exchange rate transmission	Complete exchange rate transmission	Median lag length	Import dependency (%)	Observations	Adj. R2	DW
<i>Maize</i>										
Dominican Rep.	−0.32*** (0.073)	0.39*** (0.074)	0.02 (0.126)		‡	1	99	60	0.24	1.84
Niger	−0.10** (0.045)	0.02 (0.030)	0.40*** (0.118)	†	‡		91	84	0.27	2.05
Peru	−0.21** (0.085)	0.38*** (0.043)	0.37** (0.142)	†	‡	<1	53	131	0.25	2.02
El Salvador	−0.11*** (0.030)	0.68*** (0.030)	0.47*** (0.175)	†	‡	7	47	124	0.36	1.95
Honduras	−0.17*** (0.044)	0.31*** (0.042)	−0.25*** (0.081)	†	‡	4	42	97	0.43	1.95
Ecuador	−0.15** (0.065)	0.16*** (0.021)	−2.59*** (0.041)			5	39	65	0.47	2.00
Guatemala	−0.15** (0.060)	0.17*** (0.021)	−2.59*** (0.042)			<5	39	66	0.49	2.06
Nicaragua	−0.23*** (0.067)	0.20*** (0.056)	−0.09** (0.042)	†	‡	2	14	136	0.15	1.96
Mozambique	−0.18*** (0.068)	0.34*** (0.067)	−0.07 (0.054)	†	‡	4	11	108	0.29	2.00
Chad	−0.15*** (0.040)	0.27*** (0.027)	−0.27*** (0.067)	†	‡	5	10	109	0.46	1.87
Zambia	−0.11*** (0.037)	0.05 (0.041)	0.46*** (0.035)	†	‡		8	105	0.39	2.03
Haiti	−0.10** (0.045)	−0.02 (0.035)	2.38*** (0.098)				6	73	0.08	2.00
Kenya	−0.07** (0.036)	0.19*** (0.025)	−0.26*** (0.019)	†	‡	12	5	122	0.18	1.98
Phillipines	−0.07** (0.027)	0.27*** (0.006)	−0.20*** (0.022)	†	‡	12	5	132	0.41	2.03
Malawi	−0.64*** (0.082)	0.69* (0.364)	3.32** (1.358)			<1	4	83	0.54	2.07
Mali	−0.16* (0.090)	0.19*** (0.056)	0.37* (0.202)		‡	4	3	84	0.09	2.00
Ethiopia	−0.03 (0.063)	7.18*** (0.476)	3.53*** (0.229)	†	‡		1	80	0.36	1.99
Burkina Faso	−0.10*** (0.045)	0.02 (0.023)	0.34* (0.150)				1	84	0.59	1.95
Benin	−0.15*** (0.044)	0.40*** (−0.044)	−0.20*** (−0.063)	†	‡	5	<1	129	0.28	2.00
<i>Sorghum</i>										
Benin	−0.31*** (0.091)	0.21** (0.084)	0.13 (0.184)		‡	1	7	134	0.35	1.94
Chad	−0.21** (0.090)	0.26*** (0.078)	0.54** (0.265)			2	4	111	0.25	1.99
Niger	−0.71*** (0.139)	−0.15 (0.106)	−0.50* (0.284)		‡		3	83	0.34	1.99
Senegal	−0.24* (0.134)	0.07 (0.047)	−1.12*** (0.167)				1	49	0.23	1.93
El Salvador	−0.52*** (0.166)	0.44*** (0.074)	−1.34** (0.568)			<1	<1	124	0.28	1.85
Ethiopia	−0.43** (0.192)	0.24* (0.146)	0.20** (0.082)	†	‡	1	<1	73	0.59	1.91
Mali	−0.17* (0.091)	−0.04 (0.035)	−0.24 (0.152)		‡		<1	82	0.23	1.97
Burkina Faso	−0.11*** (0.037)	0.12*** (0.024)	−1.37*** (0.102)			5	0	84	0.67	2.05

Note: Robust standard errors are shown in parentheses below the parameter estimates. *** (Pr ≤ 0.01); ** (Pr ≤ 0.05); * (Pr ≤ 0.10). † means $\phi = 0$ can be rejected at the 5% significance level; ‡ means $\phi = 1$ can be rejected at the 5% significance level.

for sorghum (Table 6). This finding of low transmission is consistent with the aggregate commodity results in Table 3.

The estimates for the country/commodity pairings also support our conclusion from the aggregate product results that the speed of adjustment of domestic consumer prices to new equilibrium with world prices—after a shock to the latter—is fairly slow. Of the 51 countries where we could establish a co-integrating relationship between consumer and world prices only 16—a third of the sample—adjust to a world price shock in no more than a month

(Table 7). Rice has the largest number of countries whose median lag length is at most a month.

We also find evidence of a long run relationship between the real exchange rate and domestic consumer price. Based again on statistical significance, such a relationship is indicated for 10 of 11 countries for wheat, 20 of 23 countries for rice, 17 of 19 countries for maize, and 6 of 8 countries for sorghum. As with the aggregate commodity results, we also find that transmission of exchange rate changes to domestic consumer prices is partial.

Table 6Ranking of countries by the long-run world price transmission elasticity. *Source:* Authors Calculations.

World Price transmission elasticity (%)	Rice	Wheat	Maize	Sorghum
>0.50	El Salvador Guatemala Haiti Honduras Pakistan Senegal	Ethiopia	El Salvador Malawi	
0.30–0.50	Cameroon Dominican Rep. Malawi Mozambique Nicaragua Nepal	Bangladesh Cameroon Ecuador Haiti India Pakistan Sri Lanka	Benin Dominican Rep. Honduras Mozambique Peru	El Salvador
0.20–0.30	Costa Rica Ecuador India Niger Philippines	El Salvador Peru	Chad Philippines	Benin Chad Ethiopia
0.10–0.20	Chad Mali Peru Vietnam	Nepal	Ecuador Guatemala Kenya Mali Nicaragua	Burkina Faso
0–0.10	Bangladesh Benin		Burkina Faso Haiti Niger Zambia	Mali Niger Senegal

Note: The long-run world price elasticity was considered to be no different than zero for cases where we could not reject the null for either the error correction term or the long-run elasticity.

Table 7Ranking of countries/commodity pairings by their media lag length. *Source:* Authors calculations.

Median lag length (months)	Rice	Wheat	Maize	Sorghum
<1 to 1	Chad Haiti Mali Nicaragua Niger Senegal Vietnam	Cameroon Ethiopia Sri Lanka	Dominican Rep. Malawi Peru	Benin El Salvador Ethiopia
2–4	Cameroon Costa Rica Dominican Rep. Guatemala Honduras India Malawi Mozambique Nepal Philippines	Ecuador El Salvador Haiti Nepal Pakistan	Honduras Mali Mozambique Nicaragua	Chad
>4	Ecuador El Salvador Pakistan Peru	Bangladesh India Peru	Benin Chad Ecuador El Salvador Guatemala Kenya Philippines	Burkina Faso

Note: The table only includes countries and commodities where statistical significance was obtained for both the error correct term and the long-run world price transmission elasticity.

In the bulk of country/commodity pairings, we can reject the hypotheses of both no and complete exchange rate transmission.

Recall the earlier discussion that for a number of reasons—not adjusting our price data for domestic transport, transaction, processing, and retail costs and other margins, and possible quality differences between imports and domestic output—our estimates of world price and exchange rate transmission are understated. Yet, without product and country specific information concerning

these costs and quality differences, we cannot say whether the understatement is stronger or weaker for any of the four commodities we cover.

Conclusion

Many economists and development specialists have argued that open economies and free trade enhance a country's food security,

in that importing from the world market can mitigate and level out the market shortages and price jumps that result from domestic supply shocks (such as drought). Yet, recent years have shown that open economies are vulnerable to a different type of food insecurity, that can come from importing world price shocks into their economies. Two key questions that governments face concerning the latter risk are what causes shocks to world agricultural and food prices, and to what degree are the price shocks transmitted to domestic prices.

This paper focuses on the second question. The paper's empirical results show generally developing countries' consumer markets are integrated with world markets for grain products. For wheat, rice, maize, and sorghum, domestic consumer food prices are found to have a long run co-integrating relationship with both world prices and real exchange rates. However, we also find that the transmission of changes in both world prices and real exchange rates to domestic consumer prices is not high. The average of the long run transmission elasticity for the world to domestic consumer price for the four products (using the aggregate estimate for each of the four across countries) is 0.25. Among the country/commodity pairings, only a few of the long run price transmission elasticities are greater than 0.5. The long run transmission elasticity for the real exchange rate to domestic consumer price for rice, wheat, maize, and sorghum are not only low but transmission is mostly incomplete. The results also show that the movement of domestic consumer prices to new equilibrium with world prices after a shock to the latter is relatively slow.

An issue for further research is the relationship between short and long run price transmission, and whether from a policy point of view there is a "desirable" relationship between the two that governments could aim for. Governments could try to smooth out over time the domestic effects from world price shocks, forestalling major price adjustment in the short run and allowing more in the long run.

Appendix A. Bewley's Transformation

The SEECM is directly estimated from Eq. (A.1) below.

$$\Delta p_t^d = \alpha_0 + \delta p_{t-1}^d + \lambda_0 \Delta w p_t^f + \lambda_1 w p_{t-1}^f + \theta_0 \Delta e_t + \theta_1 e_{t-1} + \varepsilon_t \quad (\text{A.1})$$

where $\delta = (\alpha_1 - 1)$; $\lambda_0 = \phi_0$; $\lambda_1 = (\phi_0 + \phi_1)$; $\theta_0 = \psi_0$; $\theta_1 = (\psi_0 + \psi_1)$. Eq. (A.1) is the ADL defined in Eq. (5), which itself is a transformation of the ADL (1,1) in Eq. (2). By rearranging the parameter estimates in (A.1), we get (A.2) below (equivalent to Eq. (7) earlier):

$$\Delta p_t^d = \alpha + \beta \Delta w p_t^f + \rho \Delta e_t + \delta(p_{t-1}^d - \gamma w p_{t-1}^f - \varphi e_{t-1}) + \varepsilon_t \quad (\text{A.2})$$

where $\beta = \lambda_0$ and $\rho = \theta_0$, and the standard errors are identical to those in Eq. (A.1).

The value of the error correction term $|\delta|$ is identical in both Eqs. (A.1) and (A.2). The standard error for $|\delta|$ estimated in (A.1) can be used to test the null hypotheses of no co-integration. However, the test also requires parameter estimates and their standard errors for the long run elasticities (LRE's) for the world price $w p_{t-1}^f$ and RER e_{t-1} . Estimates for these LRE's (as defined and measured by $\gamma = \frac{\lambda_1}{\delta}$ and $\varphi = \frac{\theta_1}{\delta}$ as mentioned earlier), are obtained directly from Eq. (A.2). However, we cannot obtain the standard errors to test the hypotheses that either γ or φ are significantly different from zero from Eq. (A.2).

To obtain consistent standard errors for the LRE's, we rely on Bewley's (1979) transformation. This requires that we estimate the equation:

$$p_t^d = \omega_0 - \omega_1 \Delta p_t^d + \pi_0 w p_t^f - \pi_1 \Delta w p_t^f + \kappa_0 e_t - \kappa_1 \Delta e_t + \mu_t \quad (\text{A.3})$$

where $\omega_0 = \eta \alpha_0$; $\omega_1 = \eta \delta$; $\pi_0 = \eta(\phi_0 + \phi_1)$; $\pi_1 = \eta \phi_0$; $\kappa_0 = \eta(\psi_0 + \psi_1)$; $\kappa_1 = \eta \psi_0$; $\mu_t = \eta \varepsilon_t$; $\eta = (\alpha_1 - 1)^{-1}$. Both π_0 and κ_0

can be shown to be exactly equivalent to the LRE's for the world price and exchange rate defined in Eq. (A.2).

For the world price, the equivalence of the LRE between π_0 and γ can be established as follows:

$$\pi_0 = \eta(\phi_0 + \phi_1) = \frac{(\phi_0 + \phi_1)}{(\alpha_1 - 1)} = \frac{\lambda_1}{\delta} = \gamma \quad (\text{A.4})$$

The equivalence of the LRE between φ and κ_0 can be established similarly:

$$\kappa_0 = \eta(\psi_0 + \psi_1) = \frac{(\psi_0 + \psi_1)}{(\alpha_1 - 1)} = \frac{\theta_1}{\delta} = \varphi \quad (\text{A.5})$$

According to De Boef and Keele (2008), because the LRE's are directly estimated in (A.3), their standard errors can also be directly obtained. However, we cannot get consistent estimates from Eq. (A.3) because of the inclusion of Δp_t^d on the right hand side (De Boef and Keele, 2008; Bewley, 1979). To obtain consistent estimates from Eq. (A.3), we first acquire predicted values for Δp_t^d from a modified version of Eq. (A.1), where the lags on the levels of the exogenous variables have been removed. We then substitute these predicted values to estimate the equation:

$$p_t^d = \omega_0 - \omega_1 \Delta \hat{p}_t^d + \pi_0 w p_t^f - \pi_1 \Delta w p_t^f + \kappa_0 e_t - \kappa_1 \Delta e_t + \mu_t \quad (\text{A.6})$$

where $\Delta \hat{p}_t^d$ are the predicted values obtained from Eq. (A.2). All remaining parameters in (A.6) are the same as in (A.3). As emphasized by both Bewley (1979) and De Boef and Keele (2008), this transformation is a statistical transformation for the sole purpose of obtaining the standard errors of the LRE's. As such, its R^2 does not tell us anything about the explanatory power of Eq. (A.2).

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